Eric Pizem

Zürich, Switzerland | +33 7 86 16 27 57 | epizem@ethz.ch
| LinkedIn | Portfolio GitHub

Education

ETH Zürich — MSc Mechanical Engineering 2025–Present
Relevant coursework: Aerodynamics I & II, Engine Systems, Structural Optimisation, Control Systems, Risk Analysis of Technical Systems.

EPFL — BSc Mechanical Engineering 2020–2025
Relevant coursework: Vibrations, Dynamics, Fluid Mechanics, Numerical Analysis, Manufacturing Processes, Product Design.

Engineering Projects

Pore-Scale CFD and Multiphase Flow — ETH Zürich
- Conducting pore-scale CFD simulations of water and isopropanol flow through sandstone using GeoChemFoam, OpenFOAM and ParaView.
- Investigating permeability, spontaneous imbibition, liquid saturation and residual gas trapping.

Structural Optimisation — ETH Zürich
- Performed FEM-based topology optimisation in MATLAB using SIMP, minimising structural compliance under volume and loading constraints.

Diesel Engine Control Systems — ETH Zürich
- Designed, tuned and experimentally tested a controller on a real diesel engine, evaluating stability, robustness and response to load disturbances.

Finite Element Analysis — EPFL
- Analysed natural frequencies, stress and thermal response of a thin-walled glass structure using FEM.

Elastic-Powered Vehicle — EPFL
- Designed, built and tested an elastic-powered vehicle, improving performance through prototype iterations and friction reduction.

Manual Seed Planter — Mechanical Design Project
- Designed and prototyped a low-cost manual seed planter, integrating soil opening, seed dispensing and closing.

Professional Experience

Brand Ambassador — Brasserie La Nébuleuse 2022–2025
Renens, Switzerland
- Represented the brewery through in-store promotions, customer events and brewery visits.

Laboratory Assistant — ALPOLE / EPFL Jul–Aug 2023
Sion, Switzerland
- Collected soil samples in Valais and contributed to sample preparation and field data collection.

Mechanical Engineering Intern — Mecachrome Jun 2018
Aubigny-sur-Nère, France
- Observed engine-component machining, manufacturing and quality-control processes.

Technical Skills

CAD / Mechanical Design: CATIA, Fusion 360

CFD / Simulation: GeoChemFoam, OpenFOAM, ParaView, Abaqus, ANSYS

Programming / Numerical Computing: MATLAB, Python, C++

Control / Experimental Tools: Simulink, LabVIEW

Other: Excel, MS Office

Languages: French (C2), English (C1), Spanish (B1), German (A1)

Additional Activities

Motorsport: Assisted a Jaguar E-Type racing team at Tour Auto and Le Mans Classic; measured brake temperatures during pit stops to assess a braking setup.

Martial Arts: 9 years of MMA and 10 years of judo; still actively training.

First Aid: PSC1, Canadian Bronze Cross, wilderness first aid training.

Outdoor Sports: Kitesurfing, freediving, skiing and hiking.